

Basic skills

Basic skills

Q1. use $d = b^2 - 4ac \leftarrow d \begin{cases} > 0 & 2 \\ = 0 & 1 \\ < 0 & 0 \text{ (2 complex)} \end{cases}$

so two real (distinct) solutions

$$(-2k)^2 - 4 \times 1 \times 4 < 0$$

A hand-drawn number line with a central vertical line. Two horizontal tick marks are drawn on either side of the center. The left tick mark is labeled with the number -2, and the right tick mark is labeled with the number 2.

Q 3. $\frac{x^2+3}{x-1} = Ax + B + \frac{C}{x-1}$

$= \frac{(Ax+B)(x-1) + C}{x-1}$

SO $\frac{x^2+3}{x-1} = \frac{Ax^2 + (B-A)x + C-B}{x-1}$

SO $A=1 \quad B-A=0 \quad C-B=3$
 $\Rightarrow B=A=1 \quad \Rightarrow C=3+1=4$

Q4. $\frac{3}{2}x - \frac{7}{4}$ ← linear part

$$\begin{array}{r} 2x+1 \overline{) 3x^2 - 2x + 10} \\ \underline{- 3x^2 + \frac{3}{2}x} \end{array}$$

$$\begin{array}{r} \overline{) -\frac{7}{2}x + 10} \\ \underline{- -\frac{7}{2}x - \frac{7}{4}} \end{array}$$

$\frac{47}{4}$ ← remainder

SO $\frac{3x^2 - 2x + 10}{2x+1} = \frac{3}{2}x - \frac{7}{4} + \frac{\frac{47}{4}}{2x+1}$

Q5. $y = \frac{6x}{8-4x} = \frac{6}{\frac{8}{x}-4} \rightarrow \frac{6}{-4} = -\frac{3}{2}$
as $x \rightarrow \infty$

Q6. $y = \frac{x \leftarrow N(x)}{x^2 - 4x - 12 \leftarrow D(x)}$

Best to factorise

$$y = \frac{x}{(x-6)(x+2)}$$

Intercepts: at $x=0$ $y=0$
for $y=0$ $x=0$ so $(0,0)$

Undefined at $(x-6)(x+2)=0 \Rightarrow x=-2, 6$

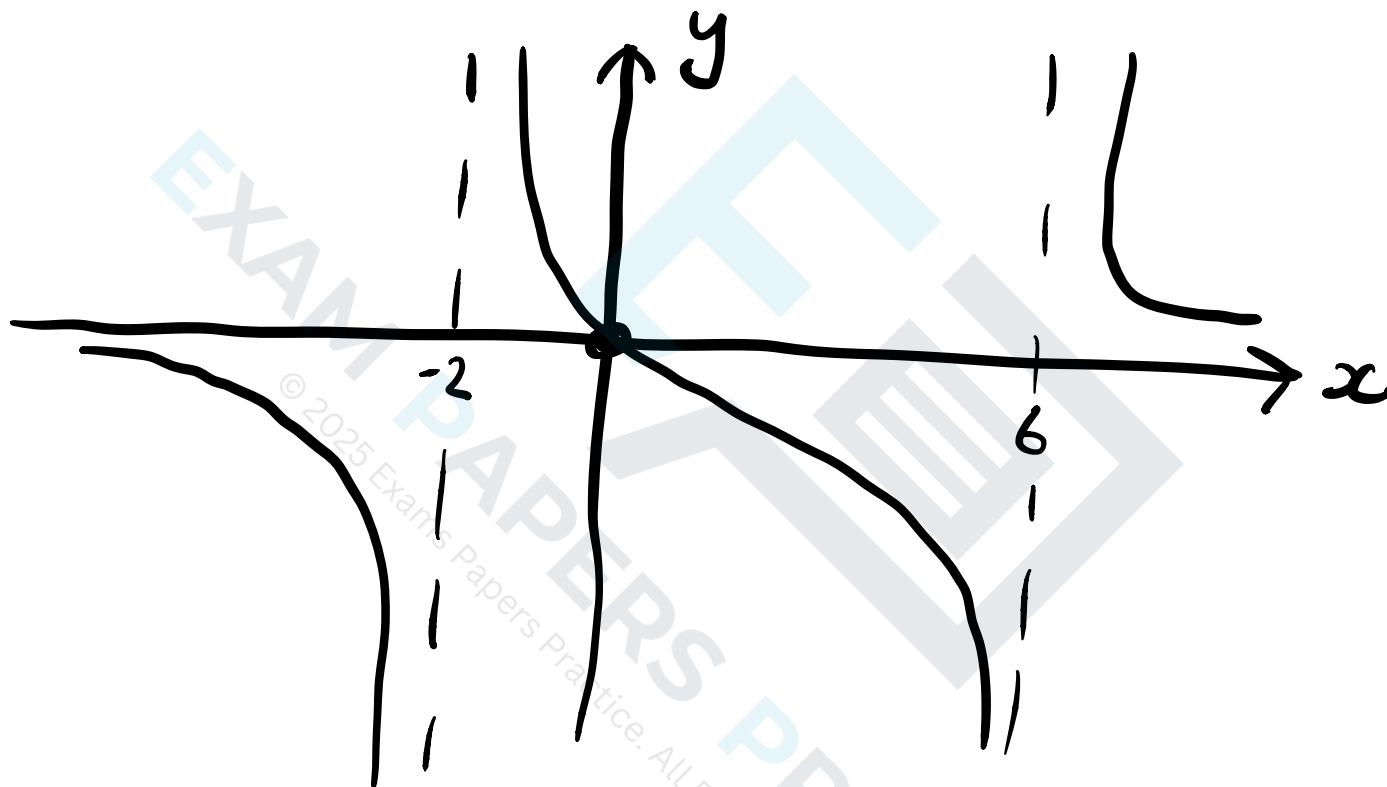
Behaviour

Around $x=-2$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{-}{-x-} = - \\ \text{Above } y \text{ is } \frac{-}{-x+} = + \end{array} \right.$

Around $x=6$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{+}{-x+} = - \\ \text{Above } y \text{ is } \frac{+}{+x+} = + \end{array} \right.$

Limit as $x \rightarrow \pm\infty$ by

$$y = \frac{\frac{1}{x}}{1 - \frac{4}{x} - \frac{12}{x^2}} \rightarrow \frac{0}{1} = 0$$



Competency skills

Q7. $y = \frac{x^2 - 2x + 3}{x + 1}$

a) $\frac{x^2 - 2x + 3}{x + 1} = Ax + B + \frac{C}{x + 1}$

$$\Rightarrow (Ax + B)(x + 1) + C = x^2 - 2x + 3$$

$$\Rightarrow Ax^2 + (A + B)x + B + C = x^2 - 2x + 3$$

so equating coeff:

$$A = 1 \quad A + B = -2 \quad B + C = 3$$

$$\Rightarrow B = -3 \quad \Rightarrow C = 6$$

$$\text{so } f(x) = x - 3 + \frac{6}{x+1}$$

b) As $x \rightarrow \infty$ $f(x) \rightarrow \infty$ along it's
oblique / asymptotic limit asymptote
of $y = x - 3$

To sketch we need other key
features:

$$\text{Intercepts: At } x=0 \quad y = \frac{3}{1} = 3$$

$$\text{for } y=0 \Rightarrow x^2 - 2x + 3 = 0$$

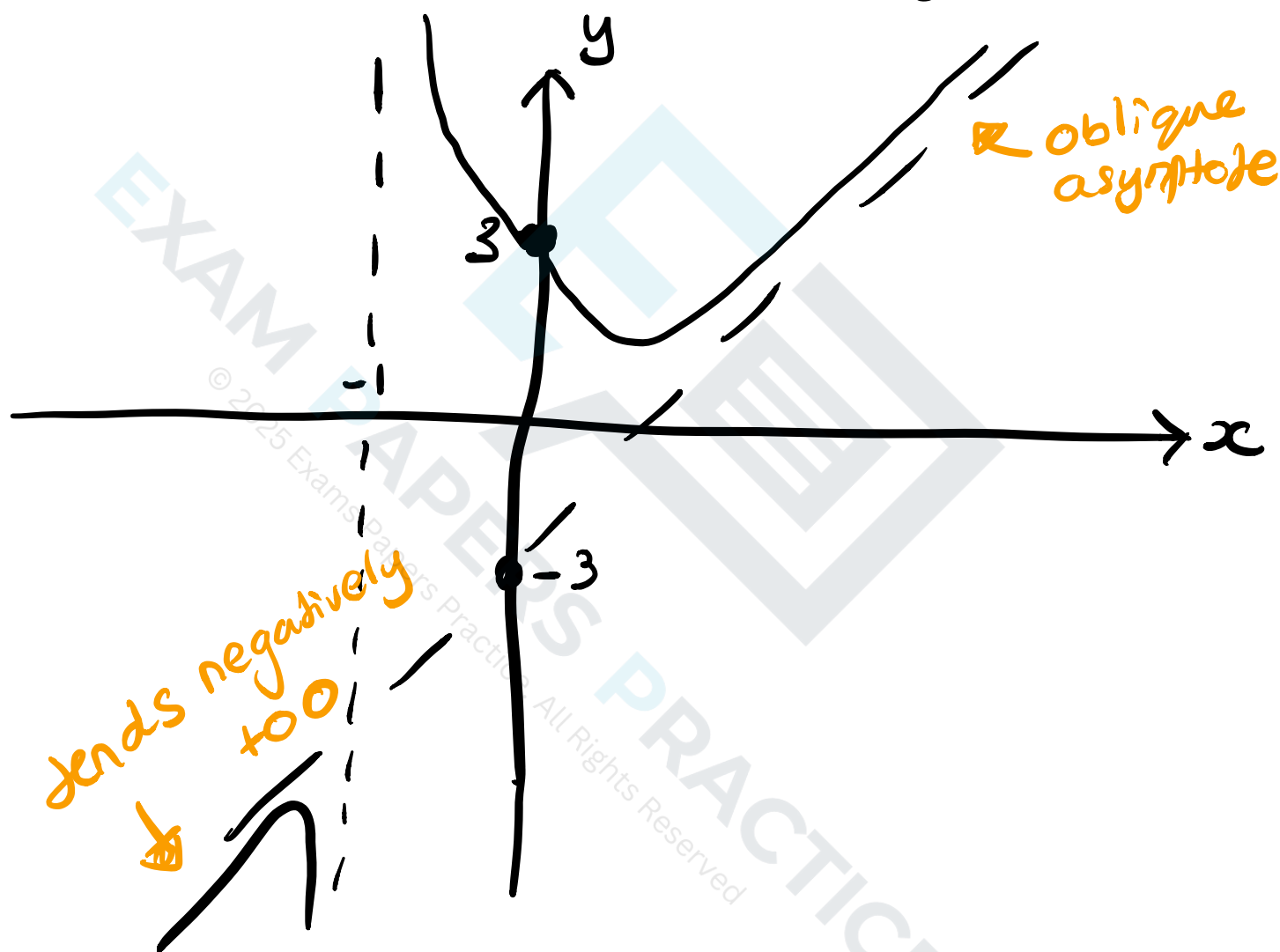
but there are no solutions

$$\text{as } d = 4 - 12 = -8 < 0$$

only
plotting
real values

undefined at $x+1=0$ $x=-1$

Around $x=-1$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{+}{-} = - \\ \text{Above } y \text{ is } \frac{+}{+} = + \end{array} \right.$



Q8. $x^2 + 5x - 1 = k(x^2 - 2x + 1)$

$$(\overset{a}{k-1})x^2 + (\overset{b}{-2k-5})x + (\overset{c}{k+1}) = 0$$

for solutions need $d \geq 0$

$$\text{so } (\overset{b^2}{-2k-5})^2 - 4(\overset{a}{k-1})(\overset{c}{k+1}) \geq 0$$

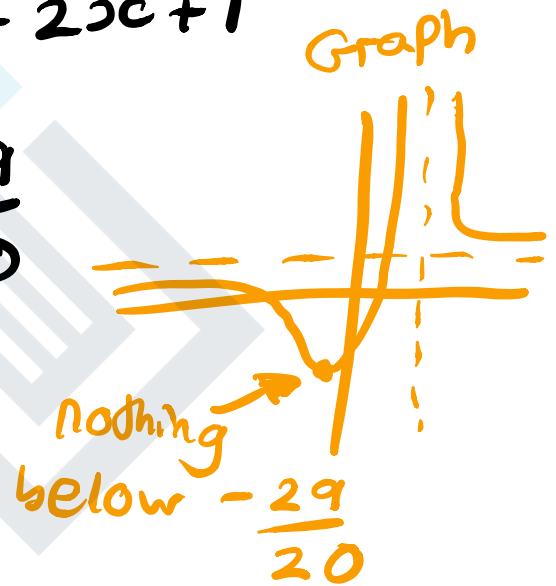
$$\Rightarrow 4k^2 + 20k + 25 - 4(k^2 - 1) \geq 0$$

$$\Rightarrow 20k + 29 \geq 0 \Rightarrow k \geq -\frac{29}{20}$$

So range of $y = \frac{x^2 + 5x - 1}{x^2 - 2x + 1}$

is

$$y \geq -\frac{29}{20}$$



Q9. $y = \frac{x^3 + 6x}{x^2 - 2}$

Intercepts: At $x=0$ $y=0$

$$\text{For } y=0 \Rightarrow x^3 + 6x = 0$$

so only $(0,0)$

$$\Rightarrow x(x^2 + 6) = 0$$

gives $x=0$

Undefined at $x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2}$

Around $x = -\sqrt{2}$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{-}{+} = - \\ \text{Above } y \text{ is } \frac{-}{-} = + \end{array} \right.$
 Around $x = \sqrt{2}$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{+}{-} = - \\ \text{Above } y \text{ is } \frac{+}{+} = + \end{array} \right.$

Limit as $x \rightarrow \pm \infty$ is

$$y = \frac{x + \frac{6}{x}}{1 - \frac{2}{x}} \rightarrow \infty$$

For oblique asymptote do division

$$\begin{array}{r}
 x^2 - 2 \quad \overline{) \quad x^3 \quad + \quad 6x} \\
 \underline{-x^3 \quad - \quad 2x} \\
 8x
 \end{array}$$

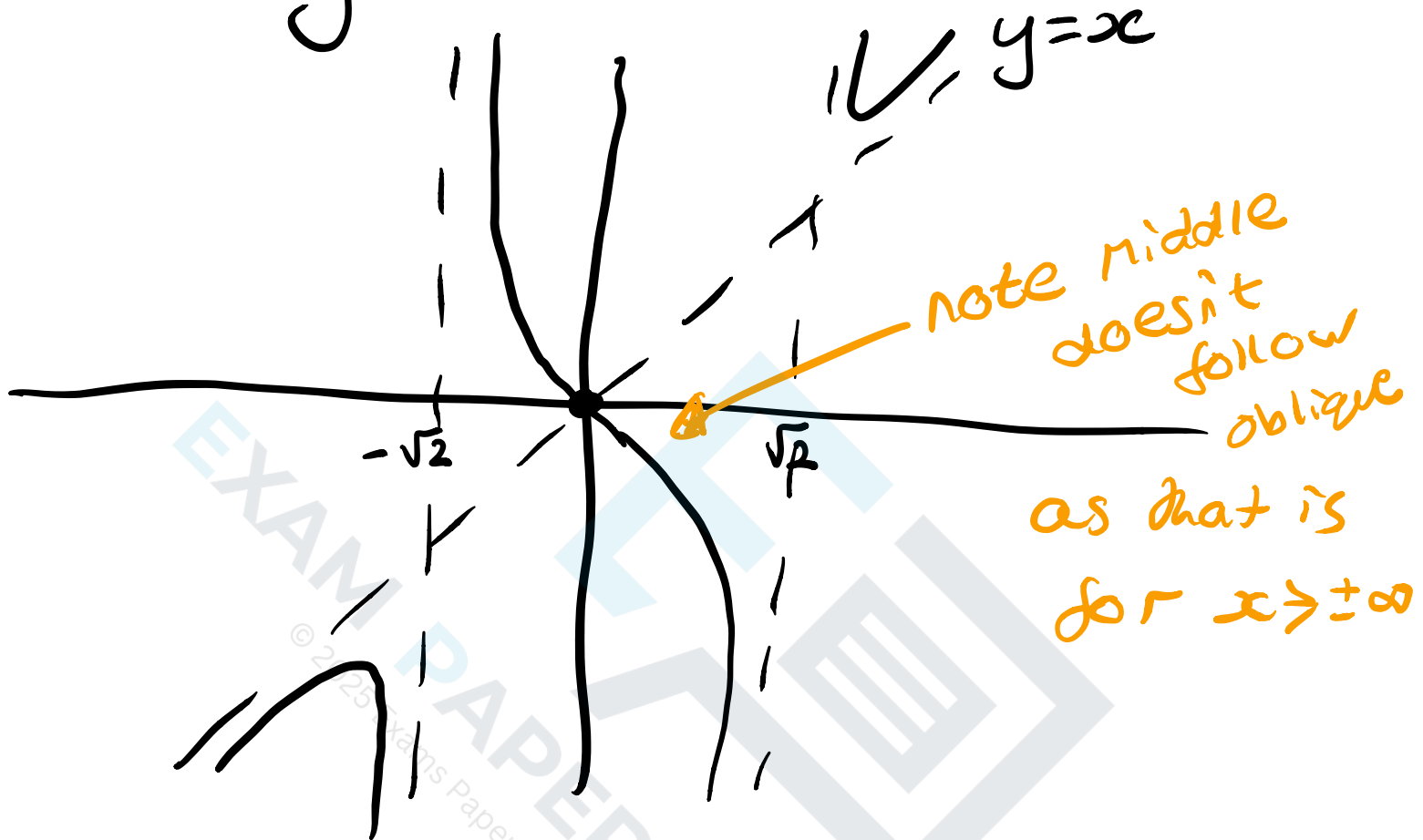
$\xrightarrow{x \cdot x}$

so

$$y = x + \frac{8x}{x^2 - 2}$$

$\swarrow$ whole $\searrow$ remainder

So $y=x$ is the oblique.



$$\text{Q10. } y = \frac{x^2 - 7x + 10}{x^2 + 10x + 21} = \frac{(x-5)(x-2)}{(x+7)(x+3)}$$

Intercepts: At $x=0$ $y = \frac{10}{21}$

$$\text{for } y=0 \Rightarrow (x-5)(x-2)=0$$

$$\Rightarrow x = 2, 5$$

undefined at $(x+7)(x+3)=0$

$$\Rightarrow x = -7, -3$$

Around $x = -7$ $\begin{cases} \text{Below } y \text{ is } \frac{-x-}{-x-} = + \\ \text{Above } y \text{ is } \frac{-x-}{+x-} = - \end{cases}$

Around $x = -3$ $\begin{cases} \text{Below } y \text{ is } \frac{-x-}{+x-} = - \\ \text{Above } y \text{ is } \frac{-x-}{+x+} = + \end{cases}$

Limit as $x \rightarrow \pm \infty$ is

$$y = \frac{1 - \frac{7}{x} + \frac{10}{x^2}}{1 + \frac{10}{x} + \frac{21}{x^2}} \rightarrow \frac{1}{1} = 1$$

For range solve $\frac{x^2 - 7x + 10}{x^2 + 10x + 21} = k$

$$\Rightarrow x^2 - 7x + 10 = k(x^2 + 10x + 21)$$

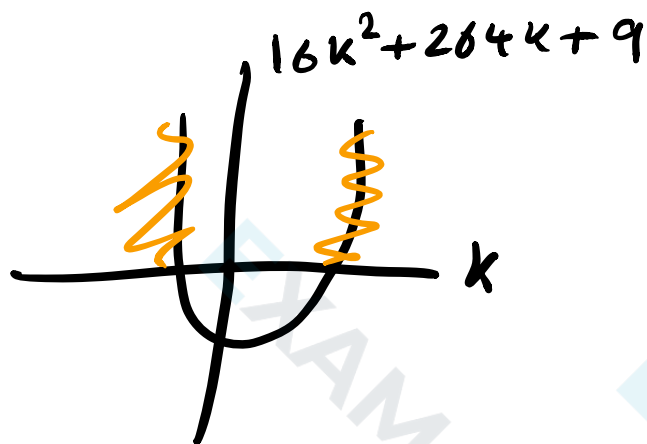
$$\Rightarrow (k-1)x^2 + (10k+7)x + (21k-10) = 0$$

So solutions when $d \geq 0$

$$(10k+7)^2 - 4(k-1)(21k-10) \geq 0$$

$$\Rightarrow 100k^2 + 140k + 49 - 4(21k^2 - 31k + 10) \geq 0$$

$$\Rightarrow 16k^2 + 264k + 9 \geq 0$$



for

$$16k^2 + 264k + 9 = 0$$

$$k = \frac{-264 \pm \sqrt{69120}}{16}$$

so

$$k \geq -\frac{33}{2} + 3\sqrt{30}$$

or

$$k \leq -\frac{33}{2} - 3\sqrt{30}$$

$$= \frac{-264 \pm 48\sqrt{30}}{16}$$

$$= -\frac{33}{2} \pm 3\sqrt{30}$$

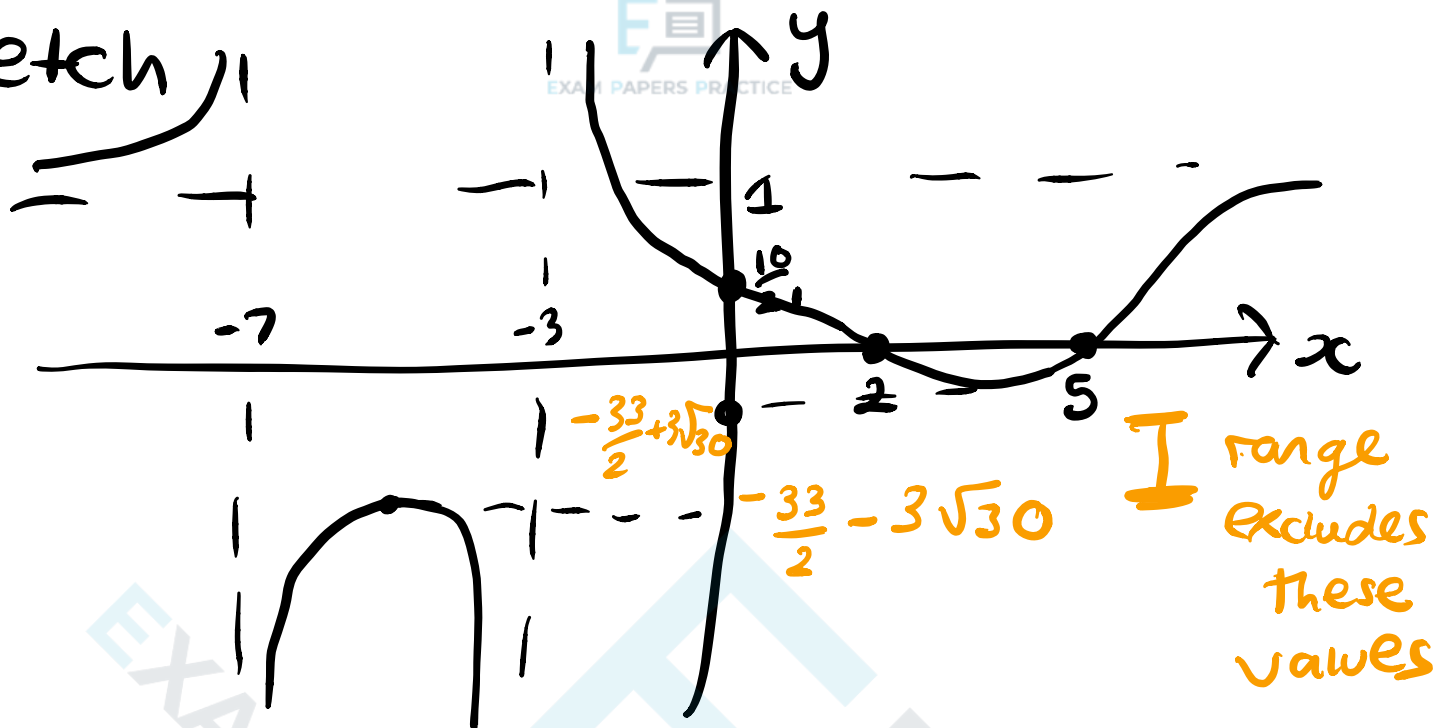
so range for y is

$$y \in \left\{ k : k \geq -\frac{33}{2} + 3\sqrt{30} \right.$$

$$\left. \text{or } k \leq -\frac{33}{2} - 3\sqrt{30} \right\}$$

$$\text{i.e. } y \geq -\frac{33}{2} + 3\sqrt{30} \text{ or } y \leq -\frac{33}{2} - 3\sqrt{30}$$

Sketch



Q11. could find range of $\frac{9x^2+8x+3}{x^2+1}$
or show

$$\frac{9x^2+8x+3}{x^2+1} \leq 11, \quad \frac{9x^2+8x+3}{x^2+1} \geq 1$$

$$9x^2+8x+3 \leq 11x^2+11$$

$$2x^2 - 8x + 8 \geq 0$$

$$2((x-2)^2 - 4) + 8 \geq 0$$

$$2(x-2)^2 \geq 0$$

yes so holds

$$9x^2+8x+3 \geq x^2+1$$

$$8x^2+8x+2 \geq 0$$

$$8\left(\left(x+\frac{1}{2}\right)^2 - \frac{1}{4}\right) + 2 \geq 0$$

$$8\left(x+\frac{1}{2}\right)^2 \geq 0$$

so yes holds

Q 12. $y = \frac{x^3}{x^2 - x - 20} = \frac{x^3}{(x-5)(x+4)}$

Intercepts : At $x=0$ $y=0$

for $y=0 \Rightarrow x^3=0 \Rightarrow x=0$

undefined at $(x-5)(x+4)=0$
 $\Rightarrow x=-4, 5$

Around $x=-4$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{-}{-x-} = - \\ \text{Above } y \text{ is } \frac{-}{-x+} = + \end{array} \right.$

$x=5$ $\left\{ \begin{array}{l} \text{Below } y \text{ is } \frac{+}{-x+} = - \\ \text{Above } y \text{ is } \frac{+}{+x+} = + \end{array} \right.$

Limit as $x \rightarrow \pm\infty$ is

$$y = \frac{x}{1 - \frac{1}{x} - \frac{20}{x^2}} \rightarrow \infty$$

Now for oblique asymptote

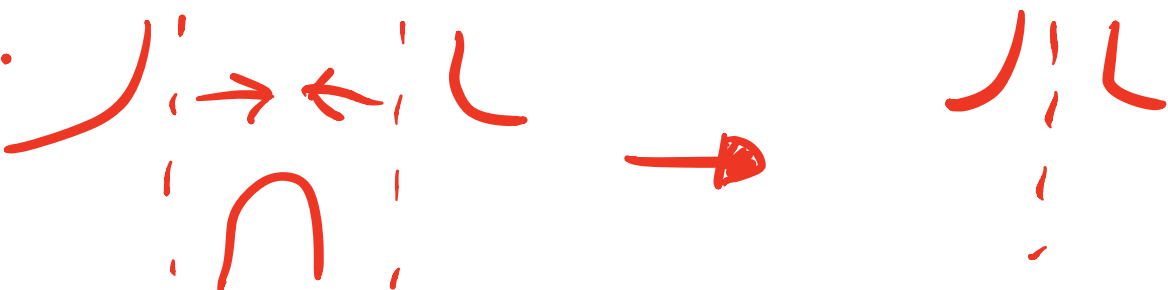
Advanced skills

Q13. For $\frac{N(x)}{D(x)} \rightarrow \infty$ as $x \rightarrow \pm\infty$

need $N(x)$ to be a higher order Polynomial than $D(x)$ so when dividing numerator still has Powers of x and denominator goes to a constant.

Q14. $y = \frac{x}{(x-a)^2}$ vertical asymptote $x=a$

special as behaviour either side is the same (due to the repeated asymptote)

e.g. 

Q15. $f(x) = \frac{x+1}{(x+2)(x+3)} \leftarrow \text{Plot } y = f(x)$

Intercepts: at $x=0$ $y = \frac{1}{2 \times 3} = \frac{1}{6}$

for $y=0 \Rightarrow x+1=0 \Rightarrow x=-1$

Undefined for $(x+2)(x+3)=0$

$\Rightarrow x = -3, -2$

Around $x = -3$ $\begin{cases} \text{Below } y \text{ is } \frac{-}{-x-} = - \\ \text{Above } y \text{ is } \frac{-}{-x+} = + \end{cases}$

$x = -2$ $\begin{cases} \text{Below } y \text{ is } \frac{-}{-x+} = + \\ \text{Above } y \text{ is } \frac{-}{+x+} = - \end{cases}$

Limit as

$y = \frac{\frac{1}{x} + \frac{1}{x^2}}{(1 + \frac{2}{x})(1 + \frac{3}{x})} \rightarrow \frac{0}{1} = 0 \text{ as } x \rightarrow \pm\infty$

For range

$\frac{x+1}{(x+2)(x+3)} = k$

$$\Rightarrow x+1 = k(x^2+5x+6)$$

$$kx^2 + (5k-1)x + (6k-1) = 0$$

real solutions for $d \geq 0$

$$\text{so } (5k-1)^2 - 4k(6k-1) \geq 0$$

$$25k^2 - 10k + 1 - 24k^2 + 4k \geq 0$$

$$k^2 - 6k + 1 \geq 0$$

$$\text{solve } k^2 - 6k + 1 = 0$$

$$k = \frac{6 \pm \sqrt{32}}{2} = 3 \pm 2\sqrt{2}$$

so

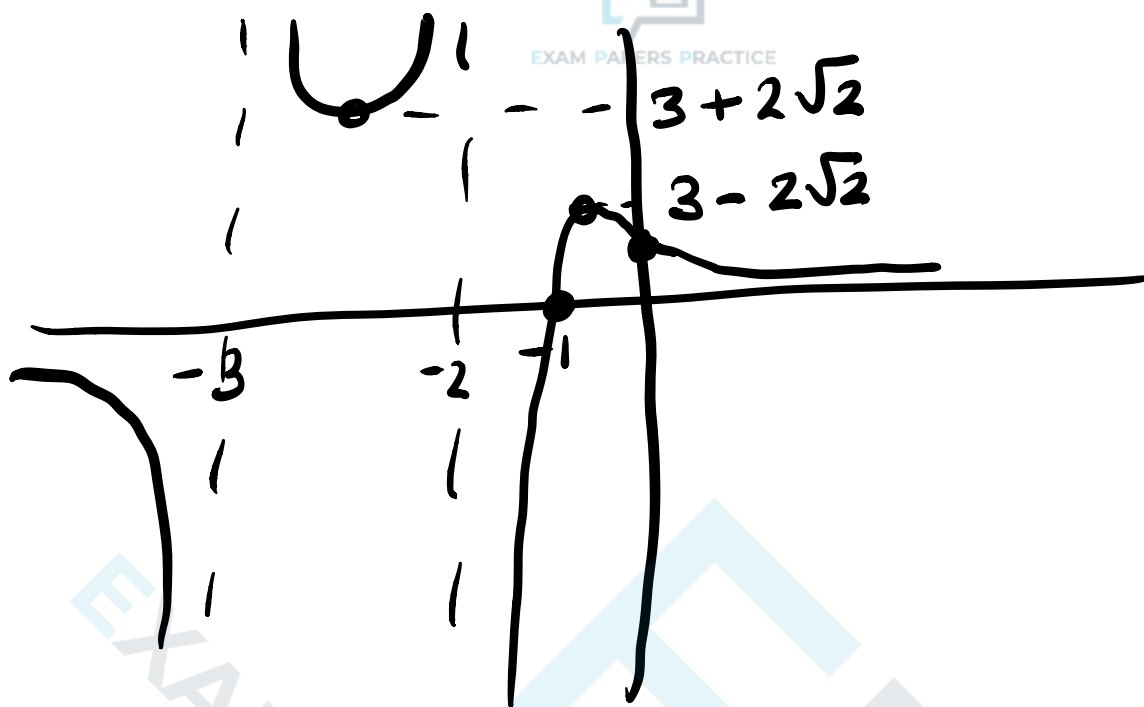


$$k \geq 3 + 2\sqrt{2}$$

or

$$k \leq 3 - 2\sqrt{2}$$

so range is $f(x) \geq 3 + 2\sqrt{2}$, $f(x) \leq 3 - 2\sqrt{2}$



So local min when $y = 3 + 2\sqrt{2}$

local max when $y = 3 - 2\sqrt{2}$

for x solve

$$\frac{x+1}{x^2+5x+6} = 3 \pm 2\sqrt{2}$$

$$\Rightarrow x+1 = (3 \pm 2\sqrt{2})(x^2+5x+6)$$

$$\Rightarrow (3 \pm 2\sqrt{2})x^2 + (14 \pm 10\sqrt{2})x + (17 \pm 12\sqrt{2}) = 0$$

Giving solution by formula

$$x = \frac{(14 \pm 10\sqrt{2}) \pm \sqrt{(14 \pm 10\sqrt{2})^2 - 4(3 \pm \sqrt{2})(17 \pm 2\sqrt{2})}}{2(3 \pm 2\sqrt{2})}$$

$$\Rightarrow x = -1 \pm \sqrt{2}$$

So local min at $(-1 - \sqrt{2}, 3 + 2\sqrt{2})$

local max at $(-1 + \sqrt{2}, 3 - 2\sqrt{2})$

Q16.

$$y = \frac{x^2 + ax + 10}{x - 5} = \frac{N(x)}{D(x)}$$

To have no oblique as $N(x)$ is higher order than $D(x)$

Need $D(x)$ to be a factor of $N(x)$

To get $x - 5$ to be a factor of $x^2 + ax + 10$

need $(x-s)(x-b) = x^2 + ax + 10$

for some a, b

$$\text{well } sb = 10$$

$$-(s+b) = a$$

$$\Rightarrow b = 2 \quad a = -7$$

Q17. $f(x) = \frac{x^4 - 5x + 2}{x^2 - 4}$ as $x \rightarrow \infty$

Clearly $f(x) \rightarrow \infty$ as

$$f(x) = \frac{x^2 - \frac{5}{x} + \frac{2}{x^2}}{1 - \frac{4}{x^2}}$$

but how? Lets do the division
to find how it goes to infinity

$$\begin{array}{r}
 x^2 + 4 \\
 \hline
 x^2 - 4 \quad \sqrt{x^4 - 5x + 2} \\
 \times x^2 \quad \swarrow \\
 \times 4 \quad \searrow \\
 \hline
 -x^4 - 4x^2 \\
 \hline
 4x^2 - 5x + 2 \\
 -4x^2 \quad \quad -16 \\
 \hline
 -5x + 18
 \end{array}$$

so $f(x) = x^2 + 4 + \frac{18 - 5x}{x^2 - 4}$

so tends to infinity along asymptotic trend

